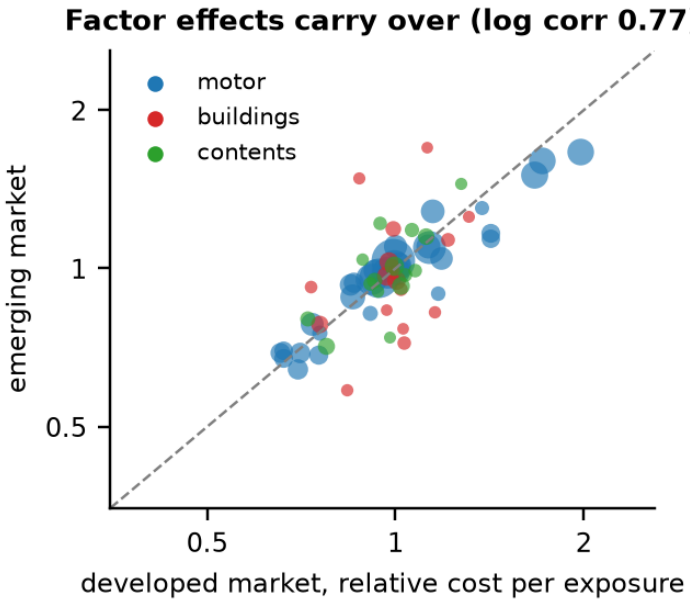
Pricing across markets with an AI-assisted workflow

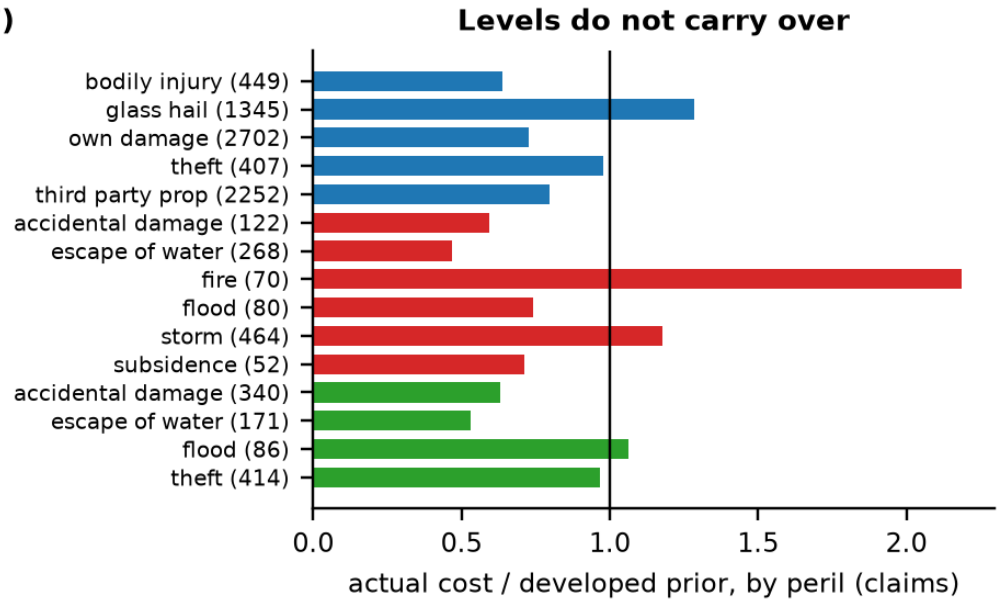
Team HS · IAA AI-Assisted Actuarial Hackathon 2026

0.13755 Part 1 private D^2, rank 8 of 51

[final] Part 2 public D^2, emerging market

What changed in the emerging market?





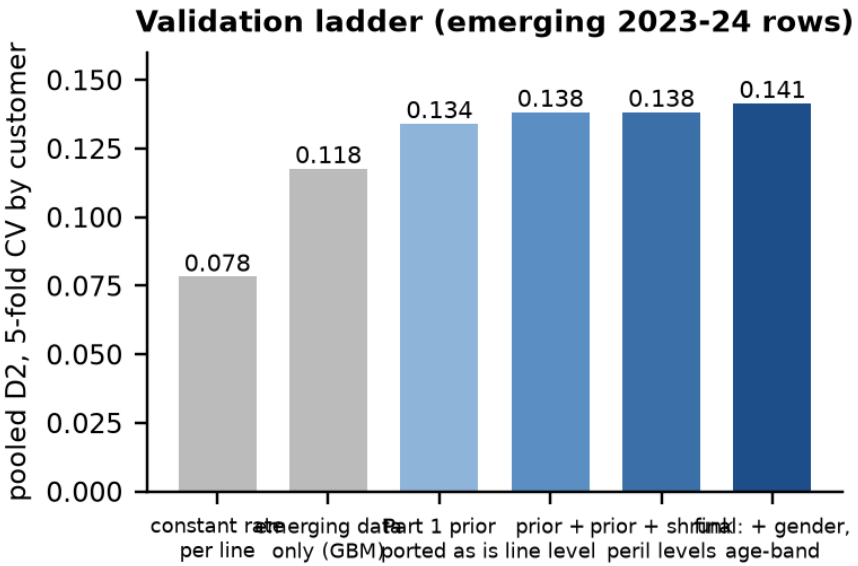
- Effects of rating factors carry over; levels do not
- Two years of data; 76% of the 2025 rows are new business

Approach: keep relativities, re-learn levels

1. **Prior**: shallow boosting per line and peril, 4 developed years, shared fields
2. **Peril levels** from the emerging data, shrunk to the line level
3. **Corrections** only if proven: robust score test, ridge GLM, nested CV
4. **Total** restored; **new-business level** calibrated (slide 5)

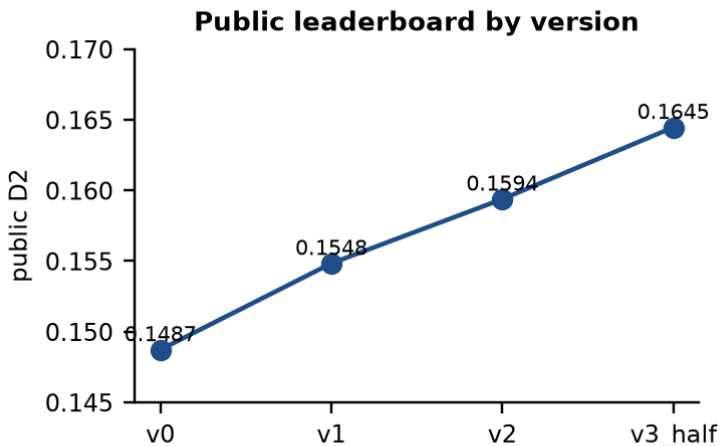
> Survived: males x1.39 on at-fault motor perils; buildings age 35-49 x0.67

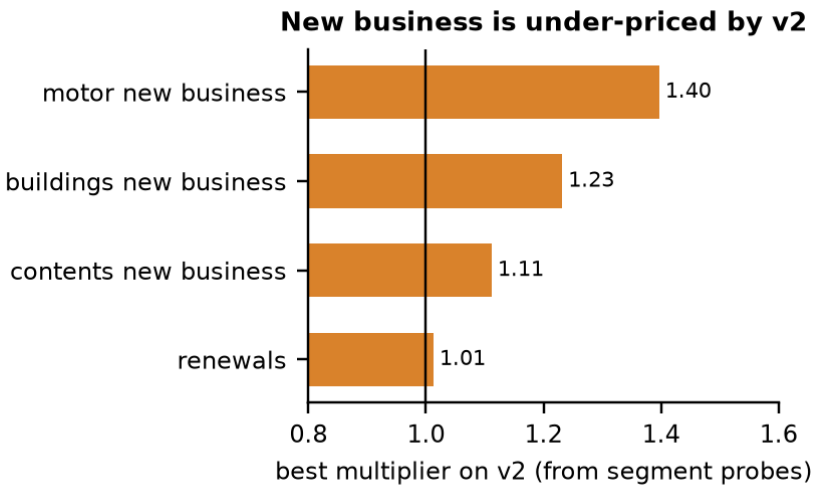
Validation ladder and guardrails



- CV grouped by customer
- 22 automated tests
- Bugs found, now tests: column order (0.036 vs 0.134), double level (+14%), timber coded T/W

New business: measured on the leaderboard





- Public score is separable by segment: one probe per segment gives its best multiplier
- Predicted 0.16433, observed 0.16446

AI-assisted workflow, verified

AI did	Humans controlled
EDA, drift, models, 40+ experiments	Adoption rule, fixed folds, nested CV
22 tests from observed failures	Tests before every submission
Report and log	Numbers copied from script output; team picks the final

Planted-effect and null tests check the AI's own code; rejected ideas stay in the log

Limits and next

- New business: largest uncertainty (broker vs direct unresolved)
- Small perils shrunk hard; 2025 weather shift not modelled
- Next: collect broker-portfolio claims early